Stromversorgung

File: power.kicad_sch

CM5

File: cm5.kicad_sch

Analogeingänge

File: analog_in.kicad_sch

Analogausgänge

File: analog_out.kicad_sch

Digital_RS485_Luefter

File: io.kicad_sch

Pumpensteuerung — AIO-Steuerplatine, Rev 0.1 (Entwurf)
Ersetzt LOGO! 8.4 + AM2 + Raspberry Pi 3B+. Kern: Raspberry Pi Compute Module 5 (eMMC, WLAN).
Versorgung 230 V AC → Mean Well IRM-30-24 → 24 V (Sensoren, XTR111, Relais) → 12 V (Lüfter) / 5 V (CM5).
Details, GPIO-Belegung und offene Punkte: hardware/SPEC.md

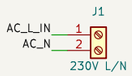
Erzeugt mit hardware/tools/gen_schematic.py — Änderungen bitte dort
Brunnensteuerung Wathlingen

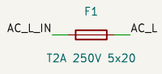
Sheet: /
File: pumpensteuerung.kicad_sch

Title: Pumpensteuerung — AIO-Steuerplatine (Übersicht)

Size: A3	Date: 2026-10-20	**Rev: 0.1**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/6

230 V AC Eingang (Netzbereich – Abstand ≥ 6 mm + Frässchlitz zur Niederspannung)

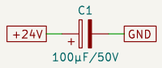


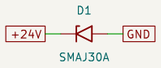




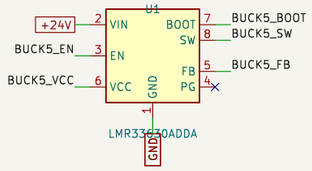


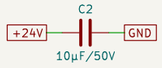
24 V Schiene



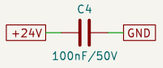


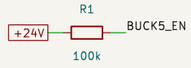
24 V → 5 V / 3 A (CM5)

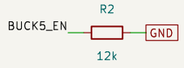




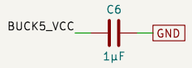


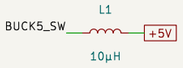








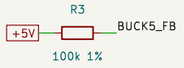


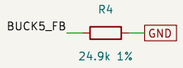




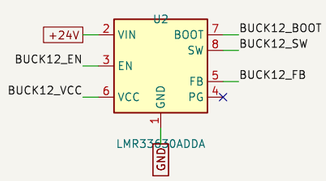








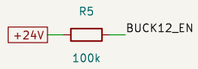
24 V → 12 V (Lüfter)

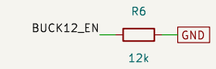


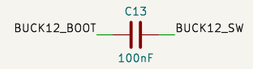




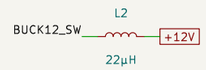


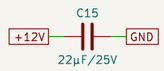


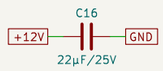


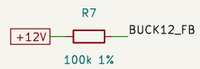






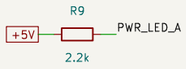








Anzeige & Power-Flags







Erzeugt mit hardware/tools/gen_schematic.py – Änderungen bitte dort

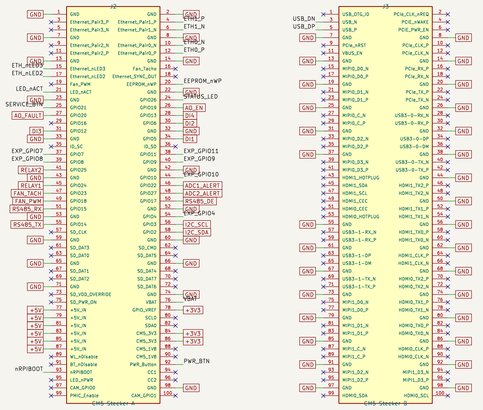
Brunnensteuerung Wathlingen

Sheet: /Stromversorgung/
File: power.kicad_sch

Title: Stromversorgung 230 V → 24 V / 12 V / 5 V

Size: A3	Date: 2026-10-20	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/6

Compute Module 5 (2× Hirose DF40C-100DS-0.4V)



Versorgung CM5

Ethernet 10/100 (MagJack, Paar 0/1)

USB-C (nur Daten: eMMC flashen per rpiboot)

Bedienung, RTC-Batterie, LEDs

I²C-Pull-ups & Erweiterungsstecker

Befestigung

Erzeugt mit hardware/tools/gen_schematic.py – Änderungen bitte dort

Brunnensteuerung Wathlingen

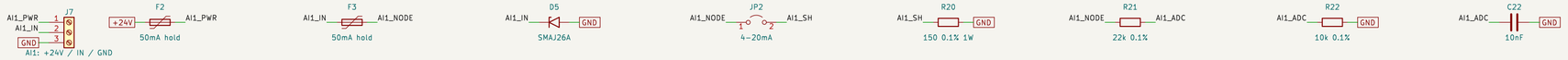
Sheet: /CM5/

File: cm5.kicad_sch

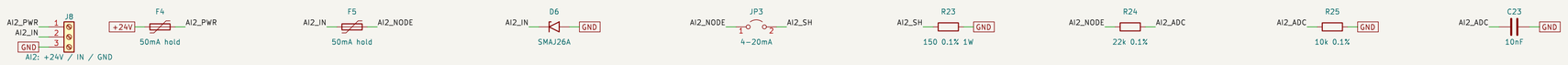
Title: Raspberry Pi Compute Module 5 + Ethernet / USB / Bedienung

Size: A1	Date: 2026-10-20	**Rev: 0.1**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 3/6

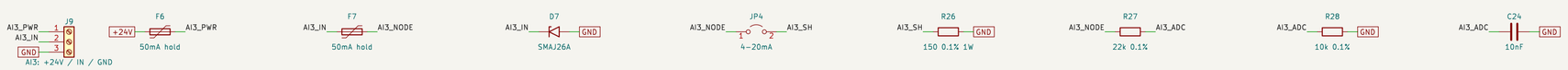
AI1 – Druck 0-10 V (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



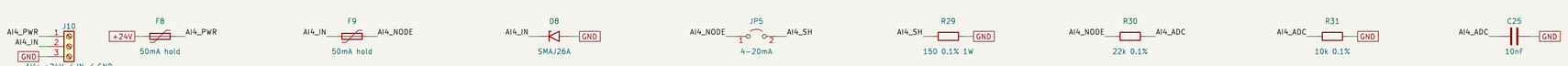
AI2 – Durchfluss 4-20 mA (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



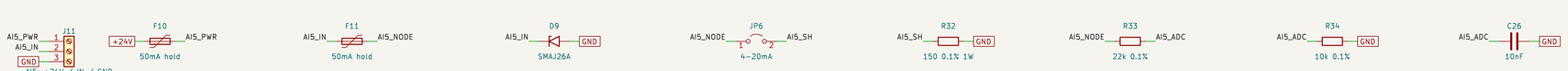
AI3 – Wassertemp 4-20 mA (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



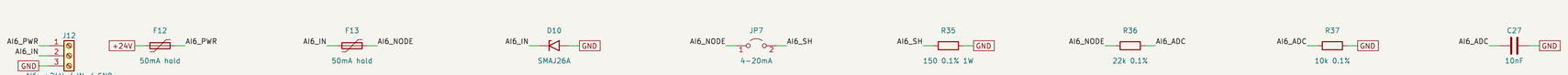
AI4 – Regentonne 4-20 mA (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



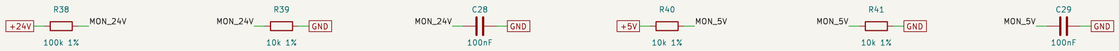
AI5 – Reserve (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



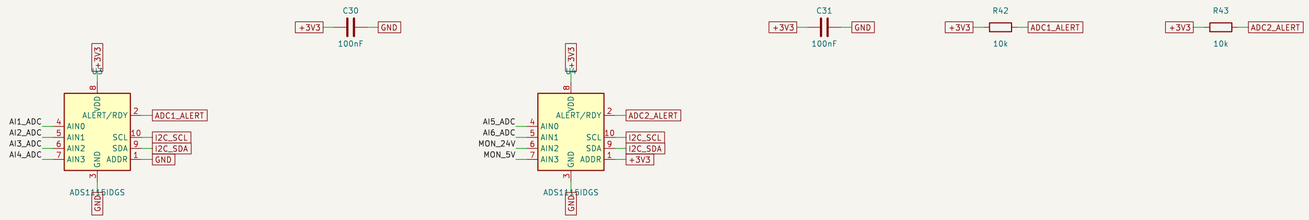
AI6 – Reserve (JP gesteckt = 4-20 mA / 150 Ω, offen = 0-10 V)



Spannungsüberwachung



ADC 1 (0x48: AI1-AI4) / ADC 2 (0x49: AI5, AI6, 24V, 5V)



Erzeugt mit hardware/tools/gen_schematic.py – Änderungen bitte dort

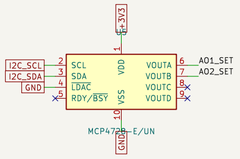
Brunnensteuerung Wathlingen

Sheet: /Analogeingänge/
File: analog_in.kicad_sch

Title: 6 × Universal-Analogeingang 0-10 V / 4-20 mA (Jumper)

Size: A2	Date: 2026-10-20	**Rev: 0.1**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 4/6

DAC (I²C 0x60, interne Ref. 2,048 V) & Ausgangsfreigabe







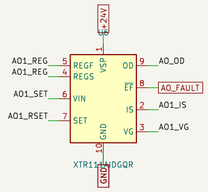








AO1 – Dosierpumpe I = 10 · U_DAC / 1 kΩ (0…20,48 mA)

















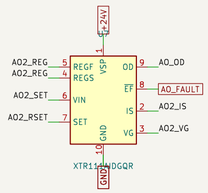








AO2 – Reserve I = 10 · U_DAC / 1 kΩ (0…20,48 mA)













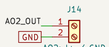












Erzeugt mit hardware/tools/gen_schematic.py – Änderungen bitte dort

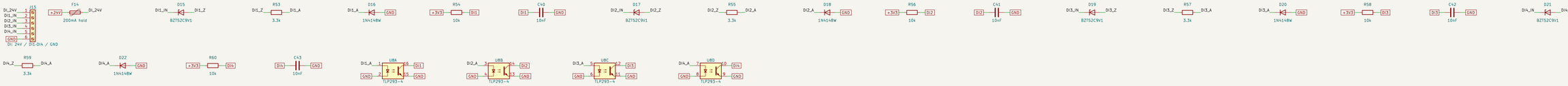
Brunnensteuerung Wathlingen

Sheet: /Analogausgänge/
File: analog_out.kicad_sch

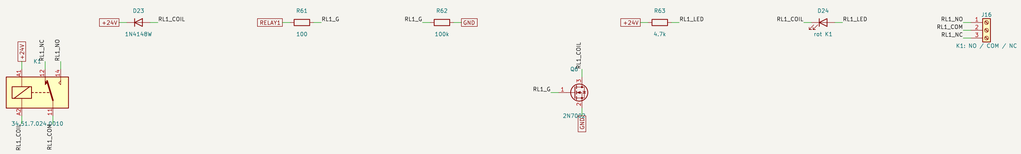
Title: 2 × Analogausgang 0/4-20 mA (Dosierpumpe + Reserve)

Size: A2	Date: 2026-10-20	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 5/6

Digitaleingänge 24 V (Schaltschwelle ≈ 12 V, aktiv = GPIO LOW)



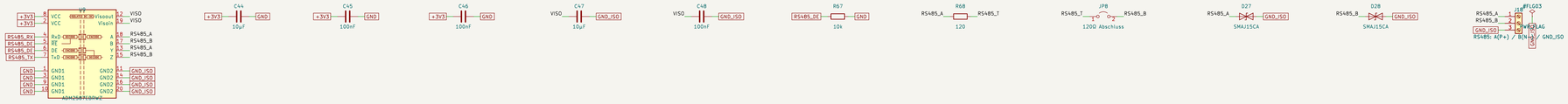
Relais K1 (230 V / 6 A, potentialfrei) – z. B. Freigabe Dosierpumpe



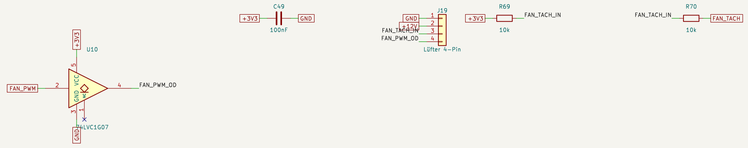
Relais K2 (230 V / 6 A, potentialfrei)



RS485 isoliert → V20 (Modbus RTU, Halbduplex, DE über RTS/GPIO17)



Gehäuselüfter 12 V, 4-Pin-PWM (GPIO18, 25 kHz) + Tacho (GPIO23)



Erzeugt mit hardware/tools/gen_schematic.py – Änderungen bitte dort
Brunnensteuerung_Wathlingen
Sheet: /Digital_RS485_Luefter/
File: lo.kicad_sch
Title: Digitaleingänge, Relais, RS485 (V20), Lüfter
Size: A1 | Date: 2026-10-20 | **Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4 | Id: 6/6